COMPETENCY STANDARD

FOR

COMPUTERISED SWEATER MACHINE OPERATION

(RMG SECTOR)

Skills for Industry Competitiveness and Innovation Program (SICIP)
Finance Division, Ministry of Finance

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INTRODUCTION:

The Skills for Industry Competitiveness and Innovation Program (SICIP) has the overall objective of developing a skilled workforce adept at handling new technologies, especially for emerging industries in Bangladesh. It will expand skills training and strengthen the development of the training ecosystem to address the skills requirements of the SICIP-selected industry sectors. The program aims to (i) increase the technology-oriented skilled workforce across emerging and priority sectors, (ii) promote inclusive skilling and upskilling opportunities for women and socially disadvantaged groups, (iii) incentivize industry-university partnerships to nurture innovation capacity and improve industry competitiveness, and (iv) foster skills for climate-resilient manufacturing processes and green technologies. The program is expected to benefit about 220,000 new and existing workers over a 6-year implementation period from 2024-2029.

The SICIP Program has, therefore, taken the initiative to enhance the employability and productivity of trainees by implementing market-responsive and job-focused training programs through public and private training providers. This will require the development of competency standards for each of the occupations/trades which will provide a structured framework in the learning process to guide training providers, ensure consistent training quality, and create an alignment between the skills provided by the training institutes and the needs of the industry.

The Competency Standard also suggests integration of YouTube or similar platforms or downloaded clips into classroom practice to ensure simulated creation of the contents so that learners are exposed to visual demonstrations before classroom instruction or practical session, which aligns with modern learning preference and supports flipped classroom models.

This competency standard is therefore developed to improve skills following the job roles and skill sets of the occupation and ensure that the required skills are aligned with industry requirements.

The document details the format, sequencing, wording, and layout of the Competency Standard for an occupation which comprises Units of Competence and its corresponding Elements.

OVERVIEW:

A Competency Standard is a written specification of the knowledge, skills, and attitudes required for the performance of a job or occupation or trade corresponding to the standard of performance required in the workplace.

Competency standard:

- provides a consistent and reliable set of components for training, recognizing, and assessing people's skills, and may also have optional support materials.
- enables industry-recognized qualifications to be awarded through direct assessment of workplace competencies
- encourages the development and delivery of flexible training that suits individual and industry requirements
- encourages learning and assessment in a work-related environment which leads to verifiable workplace outcomes.

Competency Standard has been reviewed and updated by a working group comprised of occupation-specific experts from the industry/institution, relevant consultants of SICIP.

Competency Standards describe the skills, knowledge, and attitude needed to perform effectively in the workplace. Competency Standards acknowledge that people can achieve vocational and technical competency in many ways by emphasizing what the learner can do, not how or where they learned to do it.

With Competency Standards, assessment and training may be conducted at the workplace, at training organization, during regular work, or through work experience, work placement, work simulation or any combination of these.

A Unit of Competency describes a distinct work activity that would normally be undertaken by one person in accordance with industry standards.

Units of Competency are documented in a standard format that comprises:

- Reference to Industry Sector, Occupational Title and Occupational Description
- Unit code
- Unit title
- Unit descriptor
- Unit of Competency
- Elements and performance criteria
- Variables and range statement
- Evidence guides

Together all the parts of a Unit of Competency:

- Describe a work activity
- Guide the assessor in determining whether the candidate is competent.

Identification and validation of units of competency and elements for each occupation were made by experts from in consultative workshop.

The ensuing sections of this document comprise a description of the respective occupation with all the key components of a Unit of Competence:

- An overview of all Units of Competence for the occupation and their corresponding duration required for completion of training.
- The Competency Standards that include the Unit of Competency, Unit Descriptor, Elements and Performance Criteria, Range of Variables, Curricular Content Guide, and Assessment Evidence Guide.

Units & Elements at a Glance:

Generic Competencies (20 Hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG-CSM-01-G	Apply Occupational Health and Safety (OHS) practices at workplace	1. Identify, control and report OHS hazards 2. Conduct work safely 3. Follow emergency response procedures 4. Maintain and improve health and safety in the workplace	10
SICIP-RMG- CSM -02-G	Work in a team environment	1. Identify team goals and work processes. 2. Communicate and co-operate with team members. 3. Work as a team member 4. Solve problems as a team member	10
		Total Hour	**20**

Sector Specific Competencies (30 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (Hours)
SICIP-RMG- CSM -01-S	Carry out measurements and calculations	1. Plan and prepare. 2. Obtain measurements. 3. Perform calculations.	10
SICIP-RMG- CSM -02-S	Interpret sketch and specifications in manuals	1. Identify information from manual 2. Identify sketch and specifications	10
SICIP-RMG- CSM -03-S	Apply green practices in RMG Sector	1. Interpret green concepts 2. Minimize resource use in the workplace 3. Implement waste management practices	10
		Total Hour	**30**

Occupation Specific Competencies (222 hrs.)

Code	Unit of Competency	Elements of Competency	Duration (hours)
SICIP-RMG-CSM-01-O	Understand the sweater manufacturing process	1. Identify stages of sweater manufacturing 2. Recognize different sweater types and styles 3. Identify knitting chart 4. Comprehend the role of the sweater machine in production	20
SICIP-RMG-CSM-02-O	Identify yarn and machine parts	1. Classify different types of yarn 2. Recognize main parts of the computerized sweater machine 3. Understand the role of each machine part in sweater production	20
SICIP-RMG-CSM-03-O	Understand the control system and production efficiency	1. Identify the functions of the control system 2. Interpret the tech-pack specifications for control system 3. Interpret production efficiency	24
SICIP-RMG-CSM-04-O	Set up and operate computerized sweater machine	1. Prepare the machine for operation 2. Configure the control system for the desired pattern 3. Perform pre-operation checks 4. Operate computerized sweater machine 5. Perform regular adjustments during operation	122
SICIP-RMG-CSM-05-O	Perform quality checks on knitted fabrics	1. Identify defects on knitted fabric	16

Code	Unit of Competency	Elements of Competency	Duration (hours)
		2. Inspect the finished knitted fabric	
SICIP-RMG-CSM-06-O	Perform basic machine maintenance	1. Clean the computerized sweater machine 2. Lubricate machine 3. Check machine brush, needle and jack	20
		Total Hours	**222**
		TOTAL=	**272**

COMPETENCY STANDARD: COMPUTERISED SWEATER MACHINE OPERATION

Generic Competencies:

Unit of Competency: **APPLY OCCUPATIONAL HEALTH AND SAFETY (OHS) PRACTICES AT WORKPLACE**	**Nominal Duration:** 10 Hrs**.**	**Unit Code:** SICIP-RMG-CSM-01-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to apply occupational health and safety (OHS) practices at workplace. It specifically includes the tasks of identifying, controlling and report OHS hazards, conducting work safely, following emergency response procedures and maintaining and improving health and safety in the workplace.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify, control and report OHS hazards	1.1 Work area is routinely checked for Occupational Health and Safety (OHS) hazards prior to commencing and during work. 1.2 **Hazards** are identified and corrective action is taken within the level of responsibility. 1.3 OHS hazards and incidents are reported to appropriate personnel according to workplace procedures. 1.4 **Safety signs and symbols** are identified and followed. 1.5 **Preventive control measures** are identified in accordance with OHS work standards.
2. Conduct work safely	2.1 OHS practices are applied in the workplace. 2.2 **Personal Protective Equipment (PPE)** are used.
3. Follow emergency response procedures	3.1 Emergency situations are identified and reported according to workplace requirements. 3.2 Emergency procedures are followed as appropriate to the nature of the emergency and according to workplace procedures. 3.3 **Workplace procedures** for dealing with accidents, fires and emergencies are followed whenever necessary within scope of responsibilities.
4. Maintain and improve health and safety in the workplace	4.1 Risks are identified and appropriate control measures are implemented in the workplace. 4.2 Recommendations arising from risk assessments are implemented within level of responsibility. 4.3 Safety records are maintained according to **company**

	<u>policies</u>.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Hazards	1.1. Physical Hazards- incidents, injuries, illnesses and property damage, noise, hazardous substances. 1.2. Biological Hazards- bacteria, viruses, plants, parasites, mites, molds, fungi, insects. 1.3. Chemical Hazards – dusts, fibers, mists, fumes, smoke, gasses, vapors. 1.4. Ergonomics Hazards- physically damage due to continuing same job. 1.5. Physiological factors – monotony, personal relationship, work out cycle. 1.6. Safety hazards (unsafe workplace condition) – confined space, excavations, falling objects, gas leaks, 1.7. Unsafe workers' act (Smoking in off-limited areas, Substance and alcohol abuse at work)
2. Safety signs and symbols	2.1 Direction signs (exit, emergency exit, etc.) 2.2 First aid signs 2.3 Danger Tags 2.4 Hazard signs 2.5 Safety tags 2.6 Warning signs
3. Preventive control measures	3.1 Eliminate the hazard (i.e., get rid of the dangerous machine 3.2 Isolate the hazard (i.e. keep the machine in a closed room and operate it remotely; barricade an unsafe area off) 3.3 Substitute the hazard with a safer alternative (i.e., replace the machine with a safer one) 3.4 Use administrative controls to reduce the risk (i.e. give trainings on how to use equipment safely; OHS-related topics, issue warning signage, rotation/shifting work schedule) 3.5 Use engineering controls to reduce the risk (i.e. use safety guards to machine) 3.6 Use personal protective equipment 3.7 Safety, health and work environment evaluation 3.8 Periodic and/or special medical examinations of workers
4. Personal Protective Equipment (PPE)	4.1 Apron 4.2 Safety Helmet

	4.3 Goggles 4.4 Ear plugs 4.5 Gloves 4.6 Safety Boots 4.7 Musk 4.8 Hair Net/cap
5. Workplace Procedures	5.1 Fire fighting 5.2 Medical and first aid 5.3 Evacuation 5.4 Material Safety Data Sheets (MSDSs) and manufacturers' advice
6. Company policies	6.1 Job related Standard Operating Procedures (SOPs). 6.2 Occupational Health and Safety (OHS) specific procedures.

Curricular Content Guide

1. Underpinning Knowledge	1.1. Working Area 1.2. Hazard 1.3. Safety signs and symbols 1.4. Preventive Control Measures 1.5. OHS practices 1.6. Personal Protective Equipment (PPE) 1.7. Emergency situations and emergency procedures 1.8. Workplace procedures for dealing with accidents 1.9. Risk assessments 1.10. Company policies for safety record
2. Underpinning Skills	2.1. Identifying, controlling and reporting OHS hazards, 2.2. Conducting work safely, 2.3. Following emergency response procedures 2.4. Maintaining and improving health and safety in the workplace. 2.5. Reporting hazards and risks. 2.6. Responding to emergency procedures. 2.7. Identify tools and equipment related to OHS.
3. Underpinning Attitudes	3.1. Commitment to occupational health and safety 3.2. Promptness in carrying out activities 3.3. Sincere and honest to duties 3.4. Eagerness to learn 3.5. Tidiness and timeliness 3.6. Environmental concerns 3.7. Respect for rights of peers and seniors at workplace 3.8. Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual)

	4.2 Workplace procedures 4.3 Manuals 4.4 Workplace documents, signs and symbols 4.5 Relevant specifications or work instructions. 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified, controlled and reported OHS hazards, 1.2 conducted work safely, 1.3 followed emergency response procedures 1.4 maintained and improved health and safety in the workplace. 1.5 reported hazards and risks. 1.6 responded to emergency procedures. 1.7 identified tools and equipment related to OHS
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **WORK IN A TEAM ENVIRONMENT**	**Nominal Duration:** 10 Hrs**.**	**Unit Code:** SICIP-RMG- CSM -02-G
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to work in a team environment. It specifically includes the tasks of identifying team goals and work processes, communicating and cooperating with team members, working as a team member and solving problems as a team member.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify team goals and work processes.	1.1. Team goals and collaborative decision-making processes are identified. 1.2. Role and common goals of the team are defined from available **<u>sources of information.</u>** 1.3. Team structure, responsibilities and reporting relations are identified from team discussions and other external sources.
2. Communicate and co-operate with team members.	2.1 Communication and negotiation skills are applied and maintained in all relevant situations. 2.2 Constructive contributions are made to **<u>workplace discussions</u>** on such issues as production, quality and safety. 2.3 Goals/ objectives and action plans undertaken in the workplace are communicated promptly. 2.4 Information regarding problems and issues are organized coherently to ensure clear and effective communication. 2.5 Dialogue is initiated with appropriate personnel. 2.6 Communication problems and issues are raised 2.7 Barriers to communication are identified and resolved
3. Work as a team member	3.1 Effective forms of communication are used to interact with **<u>team members</u>** in discussing team activities and objectives. 3.2 Mutual respect, empathy and active collaboration are demonstrated 3.3 Communication channels are followed as per **<u>workplace context.</u>**
4. Solve problems as a team member	4.1 Current and potential problems faced by team are identified 4.2 **<u>Problems</u>** are investigated and analyzed 4.3 Potential solutions of problem are identified 4.4 Recommendations about possible solutions are developed, documented, ranked and presented to team members for decision.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Sources of information	1.1. Organizational service rules 1.2. HR Manual 1.3. Operations Manuals 1.4. Job description 1.5. Standard operating procedures
2. Workplace discussions	2.1 Coordination meetings 2.2 Toolbox discussion 2.3 Peer-to-peer discussion
3. Team members	3.1 Coach / members 3.2 Supervisor / manager 3.3 Peers / colleagues 3.4 Other members /Employee representative of the organization.
4. Workplace context	4.1 Standard operating procedures 4.2 Workplace rules and regulations
5. Problems	5.1 Lack of communication 5.2 Unclear roles and responsibilities 5.3 Conflicts between team members 5.4 Lack of motivation/engagement 5.5 Poor time management 5.6 Overloading Team Members

Curricular Content Guide

1. Underpinning Knowledge	1.1. Sources of information 1.2. Team structure, roles, and responsibility. 1.3. Communication and negotiation skills 1.4. Workplace discussions 1.5. Effective verbal communication methods 1.6. Interpersonal communication skills. 1.7. Communication problems and issues 1.8. Barriers in communication 1.9. Malfunctions and resolutions
2. Underpinning Skills	2.1 Organizing sources of information 2.2 Identifying the role and responsibility of the team. 2.3 Identifying effective verbal communication methods 2.4 Identifying interpersonal communication skills 2.5 Applying negotiation and communication skills 2.6 Participating in team discussion. 2.7 Working as a team member. 2.8 Participating in a variety of workplace discussions

	2.9 Identifying issues 2.10 Identifying potential solutions of problem
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Eagerness to learn 3.5 Tidiness and timeliness 3.6 Environmental concerns 3.7 Respect for rights of peers and seniors at workplace 3.8 Communication with peers and seniors at workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified team goals and work processes 1.2 identified own role and responsibilities within team 1.3 communicated and cooperated with team members 1.4 working as a team member 1.5 practice problem solving within the team
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Sector-Specific Competencies (30 Hrs.)

Unit of Competency: **CARRY OUT MEASUREMENTS AND CALCULATIONS**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG-CSM-01-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for carry out measurements and calculations.It specifically includes the task of planning and preparing, obtaining measurements and performing calculations.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Plan and prepare	1.1. Work instructions are confirmed and applied to the job in hand. 1.2. Materials to be measured are identified as per job specification. 1.3. Appropriate **<u>measuring device</u>** is identified and selected based on materials to be measured. 1.4. Specifications are obtained and verified from relevant **<u>documents.</u>**
2. Obtain measurements	2.1 Method of obtaining measurement is selected and applied. 2.2 **<u>Measurements</u>** are obtained using appropriate device in accordance with job requirement. 2.3 Measurements, including area, volume, tolerance and clearance limits, are confirmed and applied
3. Perform calculations	3.1 **<u>Calculations</u>**, using basic operations, for determining material requirement are taken. 3.2 Appropriate **<u>formulas</u>** for calculating quantities are selected. 3.3 Quantities are estimated from the calculation taken. 3.4 Material quantities are calculated, confirmed and recorded within tolerances.

Range of Variables

Variable	Range May include but not limited to:
1. Measuring device	1.1 Measuring tape 1.2 Steel rule 1.3 Calculator 1.4 Sets square
2. Documents	2.1. Technical manuals 2.2. Specifications 2.3. Sketches 2.4. Drawings 2.5. Charts 2.6. Photographs
3. Measurements	3.1. Length 3.2. Width 3.3. Weight 3.4. Tolerance
4. Calculations	4.1 Addition 4.2 Subtraction 4.3 Multiplication 4.4 Division 4.5 Area 4.6 Volume 4.7 Circumference 4.8 Cubic meter 4.9 Volumetric weight
5. Formulas	5.1 Fractions 5.2 Percentages 5.3 Mixed numbers 5.4 Conversions 5.5 Scales

Curricular Content Guide

1. Underpinning Knowledge	1.1 Work instructions 1.2 Measuring devices 1.3 Specifications and documents. 1.4 Measurements 1.5 Simple calculation techniques 1.6 Appropriate formulas for calculating quantities 1.7 Estimating quantities
2. Underpinning Skills	2.1 Identifying appropriate measuring devices 2.2 Carrying out measurements 2.3 Performing calculations using appropriate formulas

3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Calculators 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 identified and selected appropriate measuring devices 1.2 carried out measurements for appropriate device 1.3 identified and selected correct mathematical formula 1.4 performed calculations using appropriate formulas
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **INTERPRET SKETCH AND SPECIFICATIONS IN MANUALS**	**Nominal Duration:** 10 Hrs.	**Unit Code:** SICIP-RMG- CSM -02-S
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to interpret sketch and specifications in manuals. It specifically includes the tasks of identifying information from manual and identifying sketch and specifications.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify information from manual	1.1. Appropriate **manuals** are collected as per sample. 1.2. Importance of manuals is recognized. 1.3. Required information are collected from manuals.
2. Identify sketch and specifications	2.1 Relevant **sketch** and **specifications** are identified. 2.2 Key **terms and abbreviations** are identified. 2.3 **Signs and symbols** are identified. 2.4 Schedules, dimensions, drawings and specifications are interpreted.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Manuals	1.1 Buyer's specification manual 1.2 Compliance manual 1.3 Maintenance procedure manual 1.4 Periodic maintenance manual 1.5 Quality manual 1.6 Signs and symbols, instruction manuals
2. Sketch	2.1 Technical sketch 2.2 Measurement sketch
3. Specifications	3.1 Product specifications 3.2 Performance specifications 3.3 Method specifications
4. Terms and abbreviations	4.1 Refers to all terms and abbreviations associated with the RMG sector
5. Signs and symbols	5.1 Include all signs and symbols associated with the RMG sector

Curricular Content Guide

1. Underpinning Knowledge	1.1 Themes on various types of RMG manuals 1.2 Sketch and specifications 1.3 Key terms and abbreviations 1.4 Signs and symbols 1.5 Rules of sketch, drawings and specifications
2. Underpinning Skills	2.1 Recognising importance of manual 2.2 Identifying relevant sketch and specifications 2.3 Identifying key terms and abbreviations 2.4 Identifying signs and symbols 2.5 Collecting information from manual as per sample 2.6 Interpreting schedules, dimensions, drawings and specifications
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in workplace 3.8 Communication with peers, sub-ordinates and seniors in workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Standard operating procedure 4.3 Workplace documents, signs and symbols 4.4 Tools, equipment and facilities appropriate to the process or activities. 4.5 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 recognized importance of manual 1.2 identified relevant sketch and specifications 1.3 identified key terms and abbreviations 1.4 identified signs and symbols 1.5 collected information from manual as per sample 1.6 interpreted schedules, dimensions, drawings and specifications
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)

3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **APPLY GREEN PRACTICES IN RMG SECTOR**	**Nominal Duration:** 10 hrs.	**Unit Code:** SICIP-RMG- CSM -03-S
Unit Descriptor: This unit covers the skills, knowledge and attitudes required for apply green practices in RMG sector. It specifically includes the tasks of interpreting green concepts, minimizing resources use in the workplace and implementing waste management practices.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Interpret green concepts	1.1 **<u>Principles of green practices in RMG (Ready-Made Garment)</u>** manufacturing are explained. 1.2 Key terms and symbols in RMG production processes are identified. 1.3 **<u>Sources of environmental impacts</u>** during RMG manufacturing activities are identified. 1.4 RMG manufacturing activities use are explained. 1.5 **<u>Ways of mitigating environmental impacts</u>** in RMG manufacturing are explained.
2. Minimize resources use in the workplace	2.1 Water, energy, and raw material consumption in RMG manufacturing are documented. 2.2 **<u>Recyclable and non-recyclable items</u>** in RMG production processes are identified. 2.3 Procedures to reduce resource consumption in RMG manufacturing are implemented. 2.4 Sustainable alternatives in RMG manufacturing are explored and applied.
3. Implement waste management practices	3.1 **<u>Different types of waste in RMG manufacturing</u>** are identified. 3.2 Hazardous waste in RMG production is disposed of according to environmental regulations. 3.3 Green habits to reduce waste in personal and professional life are practiced.

Range of Variables

Variable	Range May include but not limited to:
1. Principles of green practices in RMG (Ready-Made Garment)	1.1 Reducing energy consumption in RMG manufacturing processes. 1.2 6R for waste management in RMG production: 1.2.1 Refuse unnecessary materials and processes. 1.2.2 Reduce resource consumption and waste generation. 1.2.3 Reuse materials within the production process. 1.2.4 Recycle materials for future use in RMG manufacturing. 1.2.5 Recover usable resources from waste. 1.2.6 Repair damaged products or materials instead of discarding them. 1.3 Use of sustainable materials in RMG production with low environmental impact. 1.4 Recycling of materials used in the RMG manufacturing process. 1.5 Sustainable transportation methods for moving goods and materials within the RMG sector.
2. Sources of environmental impacts	2.1 Air Pollution in RMG Manufacturing 2.2 Water Pollution in RMG Manufacturing 2.3 Soil Degradation in RMG Manufacturing 2.4 Noise Pollution in RMG Manufacturing 2.5 Waste Generation in RMG Manufacturing 2.6 Resource Depletion in RMG Manufacturing
3. Ways of mitigating environmental impacts	3.1 Utilizing energy-efficient equipment in RMG production processes. 3.2 Adopting renewable energy sources for RMG manufacturing operations. 3.3 Implementing site protection measures to prevent pollution and reduce environmental. 3.4 Using reusable materials in production processes. 3.5 Choosing sustainable materials with low environmental impact for garment production. 3.6 Using noise-reducing equipment and scheduling production work. 3.7 Optimizing logistics and delivery to reduce transportation emissions and improve overall supply chain efficiency.
4. Recyclable and non-recyclable items	4.1 Recyclable Items in RMG Manufacturing 4.1.1 Metal components. 4.1.2 Wooden materials.

	4.1.3 Fabric scraps and textile fibers. 4.1.4 Plastic items, including buttons, tags, and packaging materials. 4.2 **Non-Recyclable Items in RMG Manufacturing** 4.2.1 Paints, coatings, and dyes. 4.2.2 Contaminated materials. 4.2.3 Treated wood and other materials.
5. Different types of waste in RMG manufacturing	5.1 Fabric waste from cutting and trimming processes. 5.2 Wood waste from packaging or display materials. 5.3 Metal waste from trims, buttons, zippers, and other hardware. 5.4 Textile waste, including scraps from garment production and defective products. 5.5 Plastic waste, such as packaging materials, labels, and plastic buttons. 5.6 Solid waste from chemical treatments or dyeing processes. 5.7 Packaging waste from raw materials and finished products.

Curricular Content Guide

1. Underpinning Knowledge	1.1 Principles of green practices in RMG (Ready-Made Garment) manufacturing. 1.2 Sources of environmental impacts. 1.3 Ways of mitigating environmental impacts 1.4 Recyclable and non-recyclable items 1.5 Different types of waste in RMG manufacturing 1.6 Hazardous waste in RMG production 1.7 Green habits to reduce waste
2. Underpinning Skills	2.1 Identifying-cyclable and non-recyclable items 2.2 Minimizing resources use in the workplace 2.3 Identifying different types of waste in RMG manufacturing 2.4 Disposing hazardous waste in RMG production 2.5 Practicing green habits to reduce waste
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided:

	4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Tools, equipment and facilities appropriate to the process or activities. 4.6 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 interpreted green concepts 1.2 minimizing resources use in the workplace 1.3 implementing waste management practices
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Occupation-Specific Competencies (222 Hrs.)

Unit of Competency: **UNDERSTAND THE SWEATER MANUFACTURING PROCESS**	**Nominal Duration:** 20 Hrs**.**	**Unit Code:** SICIP-RMG-CSM-01-O
Unit Descriptor: This unit covers the knowledge, skills and attitudes required to understand the sweater manufacturing process. it specifically includes the tasks of identifying stages of sweater manufacturing, recognizing different sweater types and styles, identifying knitting chart, comprehending the role of the sweater machine in production.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify stages of sweater manufacturing	1.1 **Manufacturing process of sweaters** is identified and key aspects of each process. 1.2 The manufacturing process breakdown is specified. 1.3 Critical components are recognized and linked to specific stages of production.
2. Recognize different sweater types and styles	2.1 **Types and styles of sweaters** are identified. 2.2 **Styles and stitches of sweaters** are recognized and described. 2.3 **Different parts of the sweater** are identified and understood in terms of their design and function. 2.4 Fabric and yarn types used in various sweater styles are recognized.
3. Identify knitting chart	3.1 **Key components** of the knitting chart are identified and explained. 3.2 Structure of the knitting chart is interpreted. 3.3 **Types of knitting charts** are recognized and described. 3.4 Special instructions on the knitting chart are identified.
4. Comprehend the role of the sweater machine in production	4.1 **Roles and responsibilities** of a computerized sweater machine operator are identified. 4.2 Relationship between machine operation and overall production efficiency is understood.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Manufacturing process of sweater	1.1. Collection of yarn 1.2. Winding 1.3. Distribution 1.4. Knitting 1.5. Inspection 1.6. Linking 1.7. Trimming 1.8. Checking 1.9. Mending 1.10. Washing 1.11. Attachment 1.12. Stitching 1.13. Quality check 1.14. Ironing 1.15. Measurement check 1.16. Tag attachment 1.17. Folding 1.18. Packing 1.19. Pre-final inspection 1.20. Final inspection 1.21. Shipment
2. Types and styles of sweater	2.1 Vest 2.2 Pullover 2.3 Cardigan
3. Styles and stiches of sweater	3.1 Styles: 3.1.1 Vest 3.1.2 Pullover 3.1.3 Cardigan 3.2 Stitches: 3.2.1 Jersey 3.2.2 Double jersey 3.2.3 Milano 3.2.4 Half cardigan 3.2.5 Full cardigan 3.2.6 Pointel 3.2.7 Mose stitch 3.2.8 Intarsia 3.2.9 Rib 3.2.10 Cable 4.3 Diamond
4. Different parts of the	4.1 Front part

sweater	4.2 Back part 4.3 Sleeve 4.4 Neck 4.5 Bottom/Cuff-rib 4.6 Pocket 4.7 Hood 4.8 Placket 4.9 Piping 4.10 Neck tape
5. Key components	5.1 Buyer 5.2 Style 5.3 Fashion 5.4 Measurement 5.5 Yarn type and composition 5.6 Description 5.7 Tension 5.8 Needle counting 5.9 Curses counting
6. Types of knitting chart	6.1 Manual 6.2 Digital
7. Roles and responsibilities	7.1 Collect order 7.2 Collect knitting chart 7.3 Collect set-up sheet 7.4 Machine set up 7.5 Operating 7.6 Check tension and design 7.7 Panel counting and bundling 7.8 Submit distribution counter 7.9 Cleaning

Curricular Content Guide

1. Underpinning Knowledge	1.1 History of the sweater 1.2 Types and styles of sweater 1.3 Styles and stiches of sweater 1.4 Parts of the sweater 1.5 Roles and responsibilities of an operator 1.6 Manufacturing process of sweater
2. Underpinning Skills	2.1 Identifying roles and responsibilities of a computerised sweater machine operator 2.2 Identifying different parts of the sweater 2.3 Identifying and describing different types and styles of sweater

	2.4 Identifying and describing the manufacturing process of sweater
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Computer/laptop/notebook 4.3 Sample sweaters 4.4 Projector 4.5 Stationary 4.6 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified different types and styles of sweater 1.2 Identified and explained the roles and responsibilities of an operator 1.3 Identified and described the manufacturing process
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: IDENTIFY YARN AND MACHINE PARTS	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-CSM-02-O

Unit Descriptor: this unit covers the knowledge, skills and attitudes required to identify yarn and machine parts. it specifically includes the tasks of classifying different types of yarn recognizing main parts of the computerized sweater machine, understanding the role of each machine part in sweater production

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **<u>bold and underlined</u>** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Classify different types of yarn	1.1. Raw materials required for sweater manufacturing are identified. 1.2. **<u>Types of yarn</u>** are identified and classified based on their fiber content and intended use. 1.3. Materials are itemized and selected according to the specific job requirements. 1.4. Yarn characteristics are determined their suitability for different sweater styles.
2. Recognize main parts of the computerized sweater machine	2.1 **<u>Type of computerized sweater machine</u>** is identified. 2.2 **<u>Machine parts</u>** are identified and checked for proper operating condition. 2.3 Machine needles are identified and understood. 2.4 Needle positioning and alignment are identified. 2.5 Other key components are identified and understood for their role in machine operation.
3. Understand the role of each machine part in sweater production	3.1 Yarn feeder is recognized for its role in controlling the flow of yarn into the machine. 3.2 Take-down unit is understood to control the speed. 3.3 Tensioning system is recognized as essential for adjusting the tension of the yarn. 3.4 Drive system is understood to power the machine. 3.5 Cam system is recognized for controlling needle movement and stitch formation.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Types of yarn	1.1 Cotton 1.2 Acrylic 1.3 Fether 1.4 Blended 1.5 Mélange 1.6 Chanel 1.7 Wool 1.8 Rayon/viscose 1.9 Cashmere 1.10 Nylon 1.11 Spandex
2. Type of computerized sweater machine	2.1 Stoll (Germany) 2.2 Shima Seiki (Japan) 2.3 Different Chinese brands
3. Machine parts	3.1 Top and side tension 3.2 Take down 3.3 Yarn feeder 3.4 Needle 3.5 Needle bed 3.6 Shrinker 3.7 Jack 3.8 Brush

Curricular Content Guide

1. Underpinning Knowledge	1.1 Raw materials for sweater 1.2 Types of yarn 1.3 Type of computerized sweater machine 1.4 Machine parts and needles 1.5 Tensioning system 1.6 Drive system and Cam system
2. Underpinning Skills	2.1 Identifying raw materials and types of yarn 2.2 Identifying type of computerized sweater machine 2.3 Identifying and checking machine parts for usability 2.4 Identifying needle positioning and alignment
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace

	3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Workplace procedures 4.3 Standard operating procedure 4.4 Workplace documents, signs and symbols 4.5 Computerized sweater machine 4.6 Tools, equipment and facilities appropriate to the process or activities. 4.7 Materials are relevant to the proposed activity.

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified raw materials and types of yarn 1.2 Identified machine parts and machine needles 1.3 Identified and checking machine parts for usability 1.4 Identified needle positioning and alignment
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **UNDERSTAND THE CONTROL SYSTEM AND PRODUCTION EFFICIENCY**	**Nominal Duration:** 24 Hrs**.**	**Unit Code:** SICIP-RMG-CSM-03-O
Unit Descriptor: this unit covers the knowledge, skills and attitudes required to understand the control system and production efficiency. it specifically includes the tasks of identifying the functions of the control system, interpreting the tech-pack specifications for control system, interpreting production efficiency.		

Elements and Performance Criteria:

(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Identify the functions of the control system	1.1. Control panel is identified. 1.2. Function keys of the control panel are identified and understood. 1.3. Error signals are identified and understood. 1.4. Indicators and displays on the control panel are recognized. 1.5. Importance of the control system is explained.
2. Interpret the tech-pack specifications for control system	1.1. **<u>Technical specifications of the control panel</u>** are identified. 1.2. Performance standards for the control system are understood and aligned with operational requirements. 1.3. Safety features are identified, ensuring compliance with safety regulations and operational safety.
3. Interpret production efficiency	3.1 Production efficiency methods are identified and described. 3.2 **<u>Production efficiency</u>** is interpreted and described. 3.3 Efficiency benchmarks are understood.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Technical specification of control panel	1.1. RPM control 1.2. Production control 1.3. Stitch length control 1.4. Fabric takes down 1.5. Racking control 1.6. Input signals 1.7. Output executors 1.8. Fabric measuring gauges (E5 - E14)
2. Production Efficiency	2.1 Machine drive time 2.2 Production output 2.3 Standard Minute Value (SMV)

Curricular Content Guide

1. Underpinning Knowledge	1.1 Control panel and function keys 1.2 Technical specifications

	1.3 Production efficiency 1.4 Efficiency benchmarks
2. Underpinning Skills	2.1 Identifying control panel 2.2 Interpreting function keys 2.3 Interpreting production efficiency
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Control panel 4.4 Technical specifications 4.5 Calculator 4.6 Projector 4.7 Stationary 4.8 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identifying control panel 1.2 Interpreting function keys 1.3 Identified and interpreted technical specifications 1.4 Interpreting production efficiency
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

SET UP AND OPERATE COMPUTERIZED SWEATER MACHINE	Duration: 122 Hrs.	SICIP-RMG-CSM-04-O
Unit Descriptor: this unit covers the knowledge, skills and attitudes required to set up and operate computerized sweater machine. it specifically includes the tasks of preparing the machine for operation, configuring the control system for the desired pattern, performing pre-operation checks, operating computerized sweater machine.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Prepare the machine for operation	1.1 **Personal Protective Equipment (PPE)** is collected and used. 1.2 Power supply and electrical connections are checked and verified for safety and proper functioning. 1.3 Machine settings are configured according to the sweater design specifications. 1.4 Program is loaded into the machine, ensuring that the correct settings. 1.5 Machine is prepared according to set-up sheet. 1.6 Yarn and thread are loaded into the machine, and the feeders are properly set. 1.7 **Safety guards** are checked and secured to prevent accidents during machine operation.
2. Configure the control system for the desired pattern	2.1 Safety protocols are checked and followed during the configuration process to ensure safe operation. 2.2 Pattern data is input into the control system. 2.3 **Control parameters** are adjusted according to the pattern's needs. 2.4 Machine interface is assessed to select the correct design program.
3. Perform pre-operation checks	3.1 Machine parts are inspected and verified to ensure they are in good working condition. 3.2 Safety equipment is checked and secured. 3.3 Yarn feed is checked to ensure it is properly threaded and aligned. 3.4 Control system settings are checked as per the desired pattern. 3.5 Electrical connections and power supply are inspected to confirm that they are stable and functioning correctly. 3.6 Test runs are conducted to verify that the machine is

	running smoothly and the settings are correct. 3.7 Needle gauge stitch density, and pattern alignment are checked.
4. Operate computerized sweater machine	4.1 Back part of the sweater is knitted according to design specifications and knitting chart. 4.2 Front part of the sweater is knitted as per the design and knitting chart. 4.3 Sleeve is knitted according to styling requirements and knitting chart. 4.4 Neck is knitted as per the design specifications and knitting chart. 4.5 Knitting process is monitored continuously during operation.
5. Perform regular adjustments during operation	5.1 Machine settings are regularly checked and adjusted. 5.2 Yarn feed is monitored. 5.3 Needle alignment and position are checked and adjusted. 5.4 Tension and stitch quality are observed during operation. 5.5 Pattern adjustments are made during operation. 5.6 Machine components are regularly checked for wear and adjusted or replaced if required.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Personal Protective Equipment (PPE)	1.1. Apron 1.2. Safety Helmet 1.3. Goggles 1.4. Ear plugs 1.5. Gloves 1.6. Safety Boots
2. Safety guards	2.1 Safety cover 2.2 Yarn cone cover
3. Control parameters	3.1 Speed 3.2 Yarn tension 3.3 Stitch density

Curricular Content Guide

1. Underpinning Knowledge	1.1 Machine settings 1.2 Safety guards 1.3 Safety protocols 1.4 Control parameters 1.5 Machine interface
2. Underpinning Skills	2.1 Configuring machine settings

	2.2 Preparing machine according to set-up sheet. 2.1 Checking safety guards 2.2 Adjusting control parameters 2.3 Checking control system settings
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual) 4.2 Personal protective equipment (PPE) 4.3 Control panel 4.4 Technical specifications 4.5 Efficiency calculation report (example) 4.6 Calculator 4.7 Projector 4.8 Stationary 4.9 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 configured machine settings 1.2 prepared machine according to set-up sheet. 1.3 checked safety guards 1.4 adjusted control parameters 1.5 checked control system settings
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency:	Nominal	Unit Code:

PERFORM QUALITY CHECKS ON KNITTED FABRICS	**Duration:** 16 Hrs**.**	SICIP-RMG-CSM-05-O
Unit Descriptor: this unit covers the knowledge, skills and attitudes required to perform quality checks on knitted fabrics. it specifically includes the tasks of identifying defects on knitted fabric, inspecting the finished knitted fabric.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	**Performance Criteria**
1. Identify defects on knitted part	1.1. **<u>Defects in the knitted part</u>** are identified. 1.2. Knitted part irregularities are recognized and noted for correction. 1.3. Thread breakage is detected and the affected areas are identified. 1.4. Machine-related defects are identified and the cause of the issue is diagnosed.
2. Inspect the finished knitted part	2.1 Visual inspections and manual checks are conducted on the finished knitted part. 2.2 Knitted part is checked for consistency in pattern, texture, and color. 2.3 Knitted part measurements are checked to ensure they conform to the required specifications.

Range of Variables

Variable	**Range** (Includes but not limited to):
1. Defects in the knitted part	1.1. Needle drops 1.2. Needle lines 1.3. Tuck or double loops 1.4. Oil spot 1.5. Dirty mark 1.6. Uneven dyeing 1.7. Spirality 1.8. Ply missing 1.9. Hole 1.10. Thick and thin yarn 1.11. Barrie effect/mistake 1.12. Laddering 1.13. Side cut 1.14. Yarn contamination

	1.15. Tension mistake 1.16. Nylon visible 1.17. Measurement defects

Curricular Content Guide

1. Underpinning Knowledge	1.1 Defects in the knitted part 1.2 Machine-related defects 1.3 Visual inspections 1.4 Knitted part
2. Underpinning Skills	2.1 Identifying defects in the knitted part 2.2 Identifying machine-related defects 2.3 Checking visual inspections 2.4 Checking knitted part
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities 3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual). 4.2 Personal protective equipment (PPE) 4.3 Tools, equipment and machinery 4.4 Knitted materials 4.5 Standard operating procedure 4.6 Projector 4.7 Stationary 4.8 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Identified defects in the knitted part 1.2 Identified machine-related defects 1.3 Checked visual inspections 1.4 Checked knitted part
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of	3.1 Competency assessment must be done in an

Assessment	assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

Unit of Competency: **PERFORM BASIC MACHINE MAINTENANCE**	Nominal Duration: 20 Hrs.	Unit Code: SICIP-RMG-CSM-06-O
Unit Descriptor: this unit covers the knowledge, skills and attitudes required to perform basic machine maintenance. it specifically includes the tasks of cleaning the computerized sweater machine, lubricating machine, checking machine brush, needle and jack.		

Elements and Performance Criteria:
(Terms in the performance criteria that are written in **bold and underlined** are elaborated in the range of variables).

Elements of Competency	Performance Criteria
1. Clean the computerized sweater machine	1.1. Machine is cleaned with an air compressor to remove dust, lint, and debris. 1.2. **<u>Cleaning tools and material</u>** are used safely and according to the manufacturer's guidelines. 1.3. Parts of the machine are wiped and cleaned with appropriate cleaning agents to ensure smooth operation. 1.4. Yarn dust is removed from the machine surface.
2. Lubricate machine	2.1 Lubrication points on the machine are identified based on the manufacturer's guidelines. 2.2 **<u>Lubrication methods</u>** are followed. 2.3 Lubricant is applied to the appropriate parts of the machine as per the machine's maintenance schedule.
3. Check machine brush, needle and jack	3.1 Machine brush is checked to ensure it is free of debris and lint, and is functioning properly. 3.2 Needles are inspected for wear, breakage, or misalignment. 3.3 Jack is checked to ensure it is operating smoothly, without obstruction or damage. 3.4 Machine brush, needles, and jack are cleaned and lubricated if needed.

Range of Variables

Variable	Range (Includes but not limited to):
1. Cleaning tools and materials	Cleaning tools: 1.1 Brushes 1.2 Cloths 1.3 Sponges 1.4 Mops 1.5 Buckets 1.6 Vacuum cleaner 1.7 Brooms and dustpans 1.8 Scrapers Cleaning materials: 1.1 Detergents 1.2 Degreasers 1.3 Solvents 1.4 Disinfectants 1.5 Lubricant cleaners 1.6 Rust removers 1.7 Glass cleaner 1.8 Polish 1.9 Soaps
2. Lubrication methods	2.1 Manual lubrication 2.2 Drip feed lubrication 2.3 Splash lubrication 2.4 Forced (pressure) lubrication 2.5 Grease pack lubrication 2.6 Mist lubrication 2.7 Oil bath lubrication 2.8 Wick feed lubrication

Curricular Content Guide

1. Underpinning Knowledge	1.1 Machine cleaning 1.2 Cleaning tools and material 1.3 Lubrication points 1.4 Manufacturer's guidelines. 1.5 Lubrication methods 1.6 Machine's maintenance schedule.
2. Underpinning Skills	2.1 Cleaning machine 2.2 Using cleaning tools and material 2.3 Identifying lubrication points 2.4 Following lubrication methods
3. Underpinning Attitudes	3.1 Commitment to occupational health and safety 3.2 Promptness in carrying out activities

	3.3 Sincere and honest to duties 3.4 Environmental concerns 3.5 Eagerness to learn 3.6 Tidiness and timeliness 3.7 Respect for rights of peers and seniors in the workplace 3.8 Communication with peers, subordinates and seniors in the workplace
4. Resource Implications	The following resources must be provided: 4.1 Workplace (simulated or actual). 4.2 Personal protective equipment (PPE) 4.3 Tools, equipment and machinery 4.4 Knitted materials 4.5 Standard operating procedure 4.6 Projector 4.7 Stationary 4.8 Learning manual

Assessment Evidence Guide

1. Critical Aspects of Competency	Assessment required evidence that the candidate: 1.1 Cleaned machine 1.2 Used cleaning tools and material 1.3 Identified lubrication points 1.4 Followed lubrication methods
2. Methods of Assessment	Competency should be assessed by: 2.1 Written test 2.2 Practical Demonstration 2.3 Oral Questioning 2.4 Portfolio (Optional)
3. Context of Assessment	3.1 Competency assessment must be done in an assessment/training center or in an actual or simulated work place after completion of the training. 3.2 Assessment should be done by a nationally certified assessor or occupation-specific industry expert.

End of the Competency Standard

Workshop/Lab Facility Standard

Course Name:	Computerised Sweater Machine Operation
Number of Trainees:	20

Course-wise Training Space (Theoretical Classroom, Workshop/ Lab/ Classroom cum Workshop):

- Classroom – 350 sft (33 sqm)
- Workshop/ lab – 800 sft (75 sqm)

  OR
- Classroom cum workshop – 1000 sft (93 sqm)

Major Training Equipment and Training Facilities:

S.N.	Major Equipment and Training facilities	Required facilities
1.	Computer/Laptops	01
2.	Multimedia projector & screen	01
3.	Internet connectivity	01
4.	Scanner	01
5.	Printer	01
6.	White board	01
7.	Weighing Scale	01
8.	Jacquard Machine (minimum)	10
9.	Wooden working table (5' x 3')	02

The following conditions must be fulfilled –

- The institute shall not use the same facilities for any other projects/organizations offering a similar course.
- The institute must provide sufficient evidence to prove ownership of the proposed training equipment.

The list denotes the minimum training equipment and facility required to effectively conduct training for a specific course. Additionally, the institute must ensure that all other necessary training tools, equipment, and furniture are available to meet the requirement of competency standards (CS) provided by SICIP.

For the operation of training course on Computerised Sweater Machine Operation the institute must ensure the availability of at least 80% of the major training equipment and training facilities (according to the CS) to be eligible for SICIP training delivery. If the score is below 80%, the remaining equipment and facilities need to be installed before the commencement of the training.

The institute will also provide all other hand tools and power tools as per CS for 20 trainees. Also, they will arrange adequate seating arrangement and classroom setup for the 20 trainees.